

CHEMNITZ, Germany

WMO: 105770

Lat: 50.7914N

Lon: 12.8719E

Elev: 1369

StdP: 13.98

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	12.5	1.1	6.1	11.7	5.9	7.8	14.7	38.2	45.5	33.7	39.9	4.3	60	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	82.3	65.2	78.7	63.7	75.6	62.4	66.8	78.4	65.1	75.7	63.6	73.1	7.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	89.8	71.0	61.2	85.0	69.3	59.6	80.2	67.9	32.1	79.2	30.7	75.9	29.5	73.3	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.0	26.9	23.4	DB	4.5	88.6	8.5	3.1	-1.6	90.9	-6.6	92.7	-11.4	94.5	-17.6	96.7
			WB	3.8	70.2	8.4	1.5	-2.3	71.3	-7.2	72.2	-11.9	73.1	-18.1	74.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	31.5	30.7	38.3	44.5	53.7	58.7	63.1	63.1	56.5	49.0	39.1	34.3
	DBStd	13.78	9.54	9.88	7.86	7.22	7.25	6.65	6.50	6.53	6.14	7.23	7.09	7.47
	HDD50	2661	572	543	368	195	46	5	0	0	9	106	330	487
	HDD65	6719	1037	961	827	614	353	210	116	115	263	497	776	950
	CDD50	1567	0	1	6	31	162	265	408	407	206	75	4	2
	CDD65	149	0	0	0	0	3	20	59	57	9	1	0	0
	CDH74	1015	0	0	0	0	20	146	414	392	38	5	0	0
	CDH80	205	0	0	0	0	0	22	88	94	1	0	0	0
Wind	WSAvg	9.5	12.3	10.2	10.7	9.0	7.7	8.0	7.2	7.8	9.0	9.8	11.1	11.5
Precipitation	PrecAvg	31.2	2.2	1.9	2.2	1.9	2.8	3.2	3.9	3.4	2.7	2.2	2.5	2.4
	PrecMax	39.9	4.3	3.6	6.0	5.0	6.6	7.3	8.6	8.9	5.1	4.0	5.4	4.3
	PrecMin	20.3	0.1	0.3	0.6	0.2	0.7	1.6	1.1	0.8	1.0	0.7	0.0	0.9
	PrecStd	4.6	1.1	0.9	1.3	1.0	1.4	1.3	1.7	1.9	1.2	0.9	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.9	56.7	63.1	69.6	76.5	83.0	86.4	87.8	78.6	71.5	58.9	54.7
		MCWB	45.5	48.0	50.3	54.4	60.6	64.7	67.8	67.0	63.7	59.4	51.0	47.1
	2%	DB	47.0	51.1	56.3	64.6	73.1	78.4	82.5	82.3	74.2	66.5	54.5	51.2
		MCWB	42.5	45.1	48.2	52.2	58.3	62.6	65.0	65.8	61.3	56.3	48.5	45.9
	5%	DB	44.5	46.5	51.5	60.3	69.7	74.3	78.6	78.2	70.4	61.7	51.4	47.6
		MCWB	41.3	42.3	44.7	50.0	56.9	61.6	63.6	64.0	59.9	53.9	47.1	43.7
	10%	DB	42.4	42.1	48.5	56.5	66.0	70.4	75.1	74.8	66.9	58.9	48.6	44.4
		MCWB	39.2	38.3	43.4	48.0	55.1	59.7	62.5	62.5	58.4	52.3	44.9	41.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	48.6	52.2	55.4	62.5	67.1	69.2	69.6	65.5	60.7	51.7	48.8
		MCDB	50.3	54.8	60.1	66.5	72.6	78.3	82.1	83.0	73.8	70.8	56.2	52.9
	2%	WB	43.5	45.5	48.2	53.1	59.9	64.7	66.9	66.9	62.9	57.3	49.5	46.3
		MCDB	46.5	50.5	55.8	62.8	70.3	74.2	78.2	79.3	71.7	63.9	53.4	50.4
	5%	WB	41.6	41.7	45.8	50.8	58.0	62.6	65.1	65.0	60.9	54.9	47.4	44.0
		MCDB	44.3	46.2	50.9	59.0	67.5	71.7	75.9	75.8	68.2	60.2	51.1	47.3
	10%	WB	39.3	38.6	43.7	48.6	56.0	60.7	63.3	63.4	59.0	53.1	45.1	41.7
		MCDB	42.2	41.7	48.2	55.3	64.7	69.4	73.3	72.5	65.2	57.7	48.4	44.3
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.5	10.6	12.9	14.8	14.1	15.5	15.2	13.1	11.6	9.1	7.5
		MCDBR	10.6	14.2	15.9	18.8	19.2	19.7	20.9	20.2	18.2	15.9	13.5	10.4
		MCWBR	8.7	10.4	10.5	11.0	10.3	9.8	10.0	9.4	9.6	9.7	9.7	8.0
	5% WB	MCDBR	9.6	13.4	14.9	17.2	17.4	17.4	18.7	18.1	16.1	13.5	12.5	9.6
		MCWBR	8.3	10.4	10.2	10.5	10.4	9.5	10.0	9.2	9.2	9.0	9.5	8.2
Clear-Sky Solar Irradiance	taub		0.298	0.320	0.362	0.401	0.404	0.402	0.417	0.410	0.375	0.349	0.321	0.293
	taud		2.392	2.358	2.288	2.226	2.251	2.282	2.244	2.282	2.360	2.420	2.437	2.404
	Ebn at Noon		233	257	264	266	270	271	265	260	257	243	221	216
	Edh at Noon		20	26	34	40	41	40	41	38	31	25	19	17
All-Sky Solar Radiation	RadAvg		249	491	798	1271	1534	1656	1572	1366	979	577	276	193
	RadStd		43	76	111	165	186	141	180	128	133	72	36	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)	+0.70	N/A	N/A	N/A	+0.56	+0.63	N/A	N/A	+104	+36

Nomenclature: See separate page